

Yan Yang

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Machine learning engineer with industry experience in deep learning, generative models, and ML embedding visualization system design. Research background in machine learning, PDE learning, user motivation/affection modeling, information retrieval, and computer vision. M.Sc. in Computer Science (Georgetown University).

SKILLS

ML / AI	PyTorch, GANs, CNNs, backpropagation, PDE learning, PINN, embedding visualization, data augmentation
Languages	Python, C++, JavaScript/TypeScript, GoLang, SQL
Systems	System design, API architecture, database schemas, caching, serialization, Kubernetes, Redis
Research	Reaction-diffusion systems, NLP, information retrieval, PDE modeling, fMRI processing/model, EEG
Tools	Git, Linux, NumPy, React, Node.js, AngularJS, FreeSurfer

EXPERIENCE

- Machine Learning R&D Engineer** | Deepcell Inc. · Mountain View, CA 2022 – 2023
- Led system design of the remodeled neural network embedding visualization system, integrated into the Cluster-Based Sorting product for real-time data exploration during imaging+sorting runs.
 - Architecture decisions spanned data storage, database schemas, API specs, serialization/deserialization, caching, parallelization, and incremental development planning.
- Research Fellow** | KAUST · Thuwal, Saudi Arabia 2023 – 2024
- Investigated data-driven methods for learning PDE parameters in reaction-diffusion systems; the underlying mathematics connects to neuroscience modeling, and in this project was applied to automate synthesis of diseased leaf images for agricultural classification models.
 - Proposed approaches to Arabic speech recognition, including acoustic source separation and spectrogram-based data augmentation.
- Independent Researcher** | Self-directed Oct 2025 – Present
- Continued KAUST-era research on reaction-diffusion system parameter learning; built a code repository with visualization and logging tools for dynamical system parameter spaces and loss landscape analysis.
 - Experimented with direct backpropagation through unrolled Gray-Scott time steps, adaptive learning rates, and alternative loss functions; evaluated PINN and surrogate model approaches.
 - Published conceptual discussion preprints on user motivation/affection modeling.
- R&D Intern** | Deepcell Inc. · Mountain View, CA 2021 – 2022
- Trained GAN models to supplement data for cell image classifiers.
 - Led development of a CNN embedding layer visualization tool used company-wide by data and bioinformatics scientists for clustering and correlation analysis.
- Member of Technical Staff** | Pure Storage · Bellevue, WA 2020 – 2021
- Implemented volume snapshot import/restore from company disks into Kubernetes.
 - Bug fixes and enhancements: multipath device issues, Python 3 migration of bash scripts; led GitHub release process for Pure Service Orchestrator v6.0.3.
- Research Assistant** | Georgetown University – InfoSense Lab · Washington, DC 2018 – 2019
- Translated reversal theory into statistical frameworks to model search engine user behavior and improve struggle detection; work published in Discover Computing (Springer Nature, 2024).
 - Revamped VR interface for behavioral data collection; provided web support for SIGIR '18 conference demonstration, acknowledged as authorship-worthy by corresponding author.

Earlier experience includes SDE roles at ColorfulClouds Tech, DataMesh, etc. (2013-2018). Full history available on request.

EDUCATION

Ph.D. studies, Computer Science University of Nevada, Reno · Reno, NV	2021 – 2022
M.Sc., Computer Science Georgetown University · Washington, DC	2017 – 2019
B.Eng. Southeast University · Nanjing, China	2002 – 2006

SELECTED PUBLICATIONS & OUTPUT

- Yang et al. (2024). Improving Searcher Struggle Detection via the Reversal Theory. Discover Computing 27(51), Springer Nature
- Yang et al. (2022). Two-Step Data Augmentation for Masked Face Detection and Recognition. IMPROVE Conference, SciTePress
- Independent research preprints and code (2025–2026) on Zenodo and GitHub, covering PDE learning & consciousness modeling

PEER REVIEW

IEEE Transactions on Affective Computing 2025 · IEEE Transactions on Neural Networks and Learning Systems 2024